

Research Summary Report

Smart Disaster Management Network Using Cisco Packet Tracer

Computer Networks Project — Theoretical Survey and Proposed Solutions

1. Abstract

Natural disasters such as floods, earthquakes, and cyclones routinely damage physical communication infrastructure at the moment it is needed most, cutting off hospitals, police, fire, and relief-coordination centres from one another. This report presents a short theoretical survey of recent research on disaster-resilient communication networks and IoT-enabled disaster management, and translates the key findings into a set of concrete, simulate-able design solutions for a Cisco Packet Tracer project. The proposed network combines a redundant physical topology, dynamic routing protocols, first-hop redundancy protocols, and VLAN-based service segmentation so that emergency services (police, fire, medical, and a central control centre) remain interconnected even when one or more links or devices fail. The report also summarises how link-failure and recovery behaviour can be demonstrated and validated inside Packet Tracer.

2. Introduction

Disaster events frequently damage cell towers, fibre links, and power infrastructure over wide areas, which fragments the networks that emergency responders depend on for coordination. Recent literature frames this problem as one of network resilience: the ability of a communications system to continue delivering an acceptable level of service, or to recover quickly, when part of the network is disrupted. A recurring theme across this literature is that resilience is not achieved through a single technology but through a combination of redundant physical paths, adaptive routing, and layered fallback mechanisms (power, radio access, and core network) working together. This project narrows that broad research area to a scope that can be realistically modelled in Cisco Packet Tracer: a wired/wireless local and inter-site network connecting emergency-service nodes through redundant links, with dynamic routing and failover so that connectivity survives the loss of any single link or device.

3. Literature Survey

3.1 Resilient Communication Infrastructure for Disaster Scenarios

A 2024 review of sustainable and resilient disaster-communication infrastructure surveys the technological building blocks needed to keep networks operating through a disaster and to recover quickly afterward, covering both communications technologies and the power systems that keep them running [1]. A related study examines the 2023 Türkiye earthquakes as a case study and proposes an integrated space-air-ground-sea network architecture that draws on high-altitude platform stations and renewable/battery-backed power sources; it distinguishes network-management strategies for the pre-disaster (preparedness) phase from those needed during the in-disaster (response) phase [2]. Other recent work models communication-network resilience mathematically, showing that architectures with multiple, distributed aggregation or data-centre nodes recover services faster after simulated disasters than networks built around a single central node [3], and surveys the broader field of network-planning, failure-modelling, and survivable-routing techniques

used to keep backbone networks available at minimum cost [4]. A 2024 study on post-earthquake radio-access-network recovery further argues that self-organizing, quickly reconfigurable radio networks will be an essential resilience feature of future (6G) networks, since disaster zones typically also see a surge in demand for the surviving capacity [5]. A more recent (2025–2026) proposal for "island-ready" cellular networks argues that individual network segments should be able to keep operating locally and independently even after being cut off from the wider core network, restoring wider connectivity once links are repaired [6]. Complementing this, a review of technological enablers for post-disaster communication highlights the value of deliberately redundant infrastructure and open, reconfigurable radio-access architectures for restoring coverage quickly after a disaster [7].

3.2 IoT- and AI-Enabled Disaster Management Systems

A second strand of literature focuses less on the transport network itself and more on the sensing and decision layer built on top of it. A 2025 paper proposes a layered architecture in which distributed IoT sensors collect real-time environmental data, an AI layer analyses it to predict floods, wildfires, or seismic events, and the results are relayed to emergency services for coordinated response [8]. A systematic literature review of IoT-based urban disaster preparedness likewise concludes that interconnected sensor and monitoring networks improve early warning and infrastructure-monitoring capability, provided the underlying communication network stays available during the event [9]. Other recent surveys describe frameworks of interconnected IoT devices for disaster management [10] and catalogue the sensor and edge-computing technologies used specifically for urban disaster monitoring [11]. Across this body of work, a consistent conclusion is that IoT- and AI-based disaster management is only as reliable as the communication backbone carrying its data — reinforcing the relevance of a resilient network layer as the foundation for any smart disaster-management system.

3.3 Redundancy and Failover Protocols at the Network Layer

A third, more implementation-focused strand of the literature deals with the specific protocols used to achieve redundancy inside a network, which map directly onto features available in Cisco Packet Tracer. Redundant mesh or ring topologies provide multiple physical paths between nodes so that a single broken link does not isolate any device [12]. At Layer 3, dynamic routing protocols such as OSPF recompute paths automatically when a link goes down, while at the default-gateway level, First Hop Redundancy Protocols such as HSRP (Cisco's Hot Standby Router Protocol) and VRRP (Virtual Router Redundancy Protocol) let a backup router transparently take over a shared virtual gateway address if the primary router fails [13][14]. Notably, a 2026 study on an enhanced VRRP framework explicitly built and tested a dual-ISP, VLAN-segmented, VRRP-based enterprise network inside Cisco Packet Tracer, demonstrating that this combination of protocols is both realistic and directly reproducible in the same simulation environment intended for this project [15]. Industry sources on redundant routing similarly emphasise that organisations such as hospitals and emergency-response centres rely on this kind of automatic failover specifically to stay connected during outages and natural disasters [16].

4. Problem Statement

During a flood, earthquake, or cyclone, one or more communication links connecting emergency-service sites (police, fire, medical, and a central disaster-control centre) may go down due to physical damage or equipment failure. A conventional, single-path network design has no way to route around such a failure,

which can isolate an emergency service exactly when coordination is most critical. The project therefore needs a network design that (a) provides more than one physical/logical path between every pair of emergency-service sites, (b) detects a link or device failure automatically, and (c) reroutes traffic within a short convergence time without requiring manual reconfiguration.

5. Proposed Solutions (Mapped to Packet Tracer Design)

Based on the survey above, the following design elements are proposed for the Packet Tracer topology. Each is a widely documented, simulate-able feature rather than a purely theoretical concept, so each can be configured and then demonstrated failing over inside the simulation.

5.1 Redundant Physical Topology

Connect the emergency-service sites (police station, fire station, hospital, and disaster-control centre) using a partial-mesh or ring topology instead of a single chain, so that at least two independent physical paths exist between any two sites. This directly reflects the topology-design principle identified in the redundancy literature, where mesh/ring layouts are highlighted as the standard way to remove single points of failure [12].

5.2 Dynamic Routing for Automatic Path Failover

Configure a dynamic routing protocol (OSPF is well supported in Packet Tracer) on the inter-site routers instead of static routes. When a link fails, OSPF routers exchange updated link-state information and recalculate the shortest available path automatically, which mirrors the network-layer redundancy strategy described in the resilience literature [13].

5.3 First Hop Redundancy at Each Site

Within each emergency-service site's LAN, deploy two multilayer switches or routers running HSRP or VRRP to share a virtual default-gateway address. If the active (master) device fails, the standby device takes over the same virtual IP/MAC address within seconds, so end devices at that site do not need any reconfiguration. This is the same configuration pattern demonstrated and tested end-to-end in Packet Tracer in the 2026 VRRP study identified in the survey [15].

5.4 VLAN Segmentation of Emergency Services

Separate each emergency service (police, fire, medical, control centre) into its own VLAN so that broadcast traffic and faults in one service's segment do not affect the others, while still allowing controlled inter-VLAN routing between them at the core layer. This follows the VLAN-segmentation practice used alongside VRRP redundancy in enterprise network designs [15].

5.5 Simulated Link/Device Failure and Recovery Testing

Validate the design inside Packet Tracer by manually disabling a link or powering down a device (simulating disaster damage) and observing that: (i) OSPF reconverges to an alternate physical path, (ii) HSRP/VRRP fails over the default gateway at the affected site, and (iii) end-to-end connectivity between emergency services is restored automatically, ideally within a few seconds. This test procedure

operationalises the "failure and recovery" evaluation approach used throughout the resilience literature [3][5].

6. How the Solutions Address the Problem

Together, the redundant topology and OSPF handle physical-path failures between sites, while HSRP/VRRP handles failures of the gateway device within a site, and VLAN segmentation contains faults to a single service rather than letting them cascade across the whole network. This layered approach mirrors the multi-mechanism resilience strategy — combining topology, routing, and gateway redundancy — that recurs across the surveyed literature rather than relying on any single fail-safe technique [1][4][16].

7. Conclusion

The reviewed literature consistently shows that disaster-resilient communication is achieved by layering multiple redundancy mechanisms rather than depending on any one component. For a Cisco Packet Tracer project, this translates into a practical and fully simulate-able design: a mesh/ring topology for physical path diversity, OSPF for automatic inter-site rerouting, HSRP/VRRP for gateway-level failover within each site, and VLAN segmentation to isolate faults. This design directly demonstrates, at a scale appropriate for a student project, the same resilience principles used in real-world post-disaster network planning.

References

- [1] B. Karaman et al., "Solutions for Sustainable and Resilient Communication Infrastructure in Disaster Relief and Management Scenarios," arXiv:2410.13977, 2024.
- [2] B. Karaman, I. Basturk, S. Taskin, F. Kara, E. Zeydan, and H. Yanikomeroglu, "Enhancing Resiliency of Integrated Space-Air-Ground-Sea Networks with Renewable Energies: A Use Case After the 2023 Türkiye Earthquake," arXiv:2405.17635, 2024.
- [3] "Resilience evaluation of communication infrastructure networks under multi-scenario disaster simulations," ScienceDirect, 2026.
- [4] "Fundamentals of Communication Networks Resilience to Disasters and Massive Disruptions," ResearchGate.
- [5] "Fast Network Recovery from Large-Scale Disasters: A Resilient and Self-Organizing RAN Framework," arXiv:2408.08609, 2024.
- [6] "Making Cellular Networks Crisis-Proof: Towards Island-Ready, Resilient-By-Design 6G Communication Network," arXiv:2512.04346.
- [7] "Five Key Enablers for Communication during and after Disasters," arXiv:2409.06822, 2024 (citing M. Matracia, N. Saeed, M. A. Kishk, and M.-S. Alouini, "Post-disaster communications: Enabling technologies, architectures, and open challenges," IEEE Open Journal of the Communications Society, vol. 3, pp. 1177–1205, 2022).
- [8] "Smart Disaster Response System using IoT and AI based Decision Making Algorithm," International Journal for Research in Applied Science & Engineering Technology (IJRASET), 2025.

- [9] "Urban Resilience through IoT-Based Disaster Preparedness and Infrastructure Monitoring: A Systematic Literature Review," ScienceDirect, 2025.
- [10] K. Sharma, D. Anand, M. Sabharwal, P. K. Tiwari, O. Cheikhrouhou, and T. Frikha, "A Disaster Management Framework Using Internet of Things-Based Interconnected Devices," Mathematical Problems in Engineering, vol. 2021, 2021.
- [11] F. Zeng, C. Pang, and H. Tang, "Sensors on the Internet of Things Systems for Urban Disaster Management: A Systematic Literature Review," Sensors, vol. 23, no. 17, p. 7475, 2023.
- [12] "Ensuring Uninterrupted Connectivity: The Crucial Role of Network Redundancy in Data Centers," DataBank, 2024.
- [13] "Communication and Networking: Building Resilient Networks — Redundancy and Failover Strategies," FasterCapital, 2025.
- [14] "Redundancy Protocols in Network Design — HSRP, VRRP," Networkers Home, 2026.
- [15] "Enhanced Virtual Router Redundancy Protocol for Optimized Network Performance and Security in Enterprise Environments," Advances in Networks, Science Publishing Group, 2026.
- [16] "What is Redundant Routing? (Failover)," Meter, 2025.